

Virtual Environment and 3D Simulation for the Visualization of Analytic Geometry and Physics

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Abstract—The Geopsics software was developed specifically to integrate Analytic Geometry and Mechanical Physics within a 3D environment, using the Three.js library for visual processing and Math.js for precise calculations. The tool enables students to perform the algebraic modeling of quadric surfaces (such as paraboloids and ellipsoids) through their canonical equations, allowing the interactive intersection of these shapes with specific planes in real time, as well as the simulation of phenomena such as uniform circular motion, in which the student inputs parameters such as mass, radius, and velocity to dynamically visualize how these phenomena occur by means of a Newtonian physics engine. Unlike general-purpose software such as Blender (excessively complex) or Desmos (limited to 2D), Geopsics focuses on the transition between the algebraic register and spatial visualization, allowing students to study a concept and deepen its visualization within a single interface.

Index Terms—Analytic Geometry, Mechanical Physics, Engineering Education, Semiotics, Geopsics, Virtual Environment.

I. INTRODUCTION

Education in engineering, and also in several programs involving the exact sciences, often depends on the union between Analytic Geometry and Mechanical Physics. From the very beginning of the course, understanding space, vectors, lines, and planes is essential for students to be able to learn more complex content. Geometry helps to apply numbers to reality and serves as a foundation for subjects such as Calculus, in which spatial perception is necessary to solve problems. At the same time, physics is fundamental because it teaches us how our reality works, explaining the motion, the space, and the time around us.

However, teaching and learning these concepts is a major challenge. When students cannot visualize how these theories work in practice, many conceptual doubts arise. The direct result of this is the high failure rates in the subjects of the early semesters, a systemic phenomenon that is well documented in the transition of incoming university students through foundational courses [1]. The researcher Raymond

Duval clearly explains this scenario by stating that we cannot touch or feel mathematical objects directly. For this reason, we need visual tools and representations in order to manipulate and understand purely abstract ideas [2].

In an attempt to solve this visualization problem, several applications and websites already exist, yet it is noticeable that there is room for new proposals in the field. Among the available options, some have high minimum requirements to run or demand a long learning curve, causing students to spend more time trying to learn how to operate the tool than studying the subject itself. Furthermore, there are other, simpler options that may be limited in their simulations because they remain confined to theory and lack the flexibility to illustrate a specific proposition based on the data provided.

In this regard, the researchers Alexandre Medeiros and Cleide Faria de Medeiros [3], in their classic analysis of the possibilities and limitations of computer simulations in Physics teaching, warn that the uncritical use of these tools may induce misconceptions if the conceptual models suggest a “black box” behavior, hiding the equations and the limits of the model from the user. The work of Medeiros and Medeiros [3] served as a true theoretical compass for our proposal, defining the pedagogical requirements necessary for the virtual transposition to be effectively oriented toward meaningful learning.

In light of the above, we began the development of Geopsics. Geopsics was conceived to be an interactive virtual environment that brings together geometry and physics on a single platform. We grounded the construction of our interface in the discussion presented by Medeiros and Medeiros [3] regarding model transparency, ensuring that the algebraic steps remain visible to the student. The major differential of the software is to offer a free creation space, in which students can change values and watch the mathematical and physical results unfold on the screen in real time, using the same formulas they would apply on paper and with the flexibility to freely adapt

the scenario to be simulated.

The objective of this work is to present how we created this platform and to show how it works as a practical tool to make learning in the exact sciences far more accessible and clear for students.

II. THEORETICAL FOUNDATION

Undoubtedly, the understanding of geometric and algebraic phenomena, and of the dynamics of physical phenomena, depends in both cases on the mastery of the mathematical tools that enable modeling in space. Equally undoubted is the fact that any student entering one of the aforementioned undergraduate programs will face the need for this understanding not only for such curricular components, but throughout the entire undergraduate period.

The curricular component of Mechanical Physics, grounded in classical Newtonian theory, intrinsically depends on vector and geometric representation. The interaction of quantities such as Gravity, Centripetal Acceleration, and Torque cannot be related solely through their scalar values, since their directions and senses determine the causality and the behavior of the entire system.

The curricular component Analytic Geometry, in turn, aims to provide the algebraic tools for manipulations using matrices, vectors, coordinate systems, and other concepts that will be exhaustively explored by disciplines such as Calculus II and Electric Circuits. Such manipulations often exceed the limitations of the medium through which they are regularly conveyed in the classroom, such as blackboards, textbooks, and static projections that are almost always restricted to two-dimensional representations.

It is in this convergence of comprehension needs that the development of Geopsics is justified, designed to facilitate strict geometric visualization together with the deterministic laws of Classical Physics. To transpose the abstract concepts of Analytic Geometry and the phenomena of Classical Physics into the virtual environment of Geopsics, it was necessary to integrate the fundamental equations governing the vector and spatial behavior described in [4] and the static and dynamic behavior of the simulated objects described in [5], which are detailed in the next section.

DECLARATION OF GENERATIVE AI USE

The authors declare the use of the artificial intelligence tools Claude (Anthropic) and Gemini (Google) for grammatical revision, translation, and consolidation of the equations described in this text. All content was subsequently reviewed and validated by the authors, who take full responsibility for the final version of this work.

III. EQUATIONS

The analytic study of two-dimensional trajectories and curves in the virtual environment is based initially on the vector and conic equations described in [4]. The modeling of linear paths in the space $\mathbb{R}^3$ occurs through the processing of the vector equation of the line (1), while the angular relation θ between two vector directions derives from the dot product formula (2).

The analytic study in the plane and of linear trajectories is based initially on the vector equation of the line (1), which determines the spatial translation of a known point (x_0, y_0, z_0) by a direction vector (v_x, v_y, v_z) scaled by the real parameter t :

$$(x, y, z) = (x_0, y_0, z_0) + (v_x, v_y, v_z) t \quad (1)$$

The angular relation θ between two vector directions derives from the dot product (2), being weighted by the respective magnitudes of the vectors in the interval between 0 and π :

$$\theta = \arccos\left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}\right) \quad (2)$$

For two-dimensional conic curves with center or vertex at the origin of the system, the parabola symmetric to the y -axis with focal parameter p is expressed in reduced form by (3), while the planar locus of the ellipse (4) and of the hyperbola with axes contained in the plane (5) are delimited by their semi-axes a and b through their canonical forms:

$$x^2 = 4py \quad (3)$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (4)$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad (5)$$

The modeling of planar barriers and three-dimensional quadrics begins with the general equation of the plane (6), structured from an orthogonal normal vector $\vec{n} = (a, b, c)$:

$$ax + by + cz + d = 0 \quad (6)$$

The quadric surfaces that delimit closed or open volumes are represented by the canonical ellipsoid (7), which reduces to a perfect sphere if $a = b = c$, and by the hyperboloids of one sheet (8), generated by three-dimensional rotation, and of two disconnected sheets (9):

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad (7)$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \quad (8)$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \quad (9)$$

Moreover, the parabolic and conic topologies extend to the elliptic paraboloid (10), characterized by elliptic horizontal sections, to the hyperbolic paraboloid or saddle surface (11), with negative Gaussian curvature, and to the quadric cone whose vertex intercepts exactly the origin of the system (12):

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = z \quad (10)$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = z \quad (11)$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0 \quad (12)$$

The metric routines for determining absolute spatial separations are based on the direct Euclidean distance between two points in space (13), founded on the three-dimensional extension of the Pythagorean Theorem:

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \quad (13)$$

For linear and surface elements, the smallest orthogonal distance from a point to a line (14) is computed through the geometric ratio of the cross product, and the absolute distance from a point to a reference plane (15), which uses the normal vector as a normalization factor:

$$d(p, r) = \frac{|\vec{v} \times \vec{AP}|}{|\vec{v}|} \quad (14)$$

$$d(p_0, \pi) = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}} \quad (15)$$

Finally, the minimum separation between two skew straight trajectories in space (16) is defined by the quotient between the volume of the parallelepiped (scalar triple product in the numerator) and the area of its base (magnitude of the cross product in the denominator):

$$d(r_1, r_2) = \frac{|(\vec{v}_1, \vec{v}_2, A_1 \vec{A}_2)|}{|\vec{v}_1 \times \vec{v}_2|} \quad (16)$$

The Mechanical Physics Module operates independently, implementing a classical physics engine capable of computing and animating real kinematic and dynamic phenomena based on the Newtonian formalism of Halliday, Resnick, and Walker [5].

For the modeling of continuous rectilinear motion with constant acceleration, the temporal position of a particle is calculated through the equation of motion (17), where x_0 represents the initial position of the object, v_0 its initial linear velocity, and a the constant acceleration imposed along the axis of motion.

$$x(t) = x_0 + v_0 t + \frac{1}{2} a t^2 \quad (17)$$

In scenarios where the instantaneous velocity of the body must be determined without prior knowledge of the elapsed time interval, the physics engine calculates the spatial displacement using Torricelli's equation (18), a tool that establishes the relation between the final linear velocity v and the scalar displacement performed Δx .

$$v^2 = v_0^2 + 2a \Delta x \quad (18)$$

Additionally, for complex ballistic simulations in multiple dimensions, such as oblique launches and free-fall trajectories, the system processes independent parametric equations for each coordinate axis, modeling the vertical behavior (19) from the initial launch height y_0 , the vertical component of the initial velocity v_{0y} , and the uniform local gravitational acceleration g :

$$y(t) = y_0 + v_{0y} t - \frac{1}{2} g t^2 \quad (19)$$

The causal core that alters the inertial state of the simulated objects on the platform is based on the application of external

forces and vector fields, computed from Newton's Second Law (20).

$$\sum \vec{F} = m \vec{a} \quad (20)$$

This fundamental law expresses that the resultant vector force applied to a body is directly proportional to the vector acceleration $\vec{a}$ it acquires, with the constant mass m being the intrinsic property that quantifies the inertia of the simulated object.

When simulating contact interactions in three-dimensional environments, such as the sliding of blocks along ramps and textured surfaces, the engine introduces dissipative kinetic friction forces (21), whose magnitude f_k is governed by the dynamic friction coefficient μ_k configured by the user in the interface and by the intensity of the normal contact force F_N .

$$f_k = \mu_k F_N \quad (21)$$

Furthermore, integrating the geometric constraints of three-dimensional space into the dynamic behavior, the intensity of the centripetal force (22) exerted on the body along circular trajectories is calculated based on the instantaneous scalar velocity v and the spatial radius of curvature R :

$$F_c = \frac{m \cdot v^2}{R} \quad (22)$$

Finally, the behavior of shocks and impacts between isolated point bodies is modeled from the law of conservation of linear momentum (23), ensuring that the total linear momentum of the system remains constant by relating the masses m_1 and m_2 with their respective initial velocity vectors $(\vec{v}_{1i}, \vec{v}_{2i})$ before the impact and final ones $(\vec{v}_{1f}, \vec{v}_{2f})$ after the collision.

$$m_1 \vec{v}_{1i} + m_2 \vec{v}_{2i} = m_1 \vec{v}_{1f} + m_2 \vec{v}_{2f} \quad (23)$$

The consequent dissipation of kinetic energy during these mechanical shocks — such as in the impact against barriers or in the “bounce” of an object against the flat ground — is interactively controlled by the student through the coefficient of restitution (24). This relation determines the final separation velocity v_f of the body after colliding orthogonally against an ideal reference surface, given its approach velocity v_i , using the interface parameter e as a continuous regulating factor that varies from 0 for perfectly inelastic impacts to 1 for perfectly elastic impacts:

$$v_f = -e \cdot v_i \quad (24)$$

IV. METHODOLOGY

To address these learning barriers, we created a simulation platform that works as a bridge between basic geometry and the functioning of the laws of physics. Although well-known software such as GeoGebra, Blender, and Desmos already exists, these tools have limitations when used for this specific purpose. GeoGebra focuses on mathematical rigor and, although the literature demonstrates its value in overcoming the two-dimensional limitation in higher education [6], it can be unfriendly for students in the early stages of study, and adapting it for simultaneous three-dimensional dynamic

physical simulation lies outside its main scope; Blender is too complex for basic learning, being aimed at professional animation; and Desmos is limited mainly to 2D graphs.

The approach we decided to follow in order to contribute to the academic community was to develop a free creation environment that gives students complete freedom to experiment. This would allow concepts to be tested in practice, such as performing vector calculations and simulating the launch of objects, making the study much clearer.

The structure of the system is divided into three main parts: the visual part, which uses the Three.js library [7] to draw 3D objects and mathematical shapes in real time; the interface part, built with HTML5 and CSS to ensure that the use of the screen is simple and organized; and the processing core, built in JavaScript together with the Math.js library [8], responsible for performing all mathematical and physical calculations with precision.

To ensure that the software is a positive contribution to the educational field, we tested its operation by applying Geopsics to problems solved in reference textbooks, ensuring that the tool is useful in the learning process.

V. GEOPSICS INTERFACE AND FEATURES

A. User Interface

Below, we detail the Geopsics interface, which was designed from a cooperative and intuitive perspective, aiming to facilitate interaction with the concepts addressed. The platform presents an initial screen divided into two main modules: *Analytic Geometry* and *Physics*.



Figure 1. Geopsics home page.

Within each module, the user has dedicated control panels for experimentation. In the Analytic Geometry module, the tool allows the manipulation of objects through two complementary approaches: direct interaction, in which geometric shapes can be dragged and resized in space, or algebraic modeling, in which the user describes the curves and surfaces through their mathematical equations.

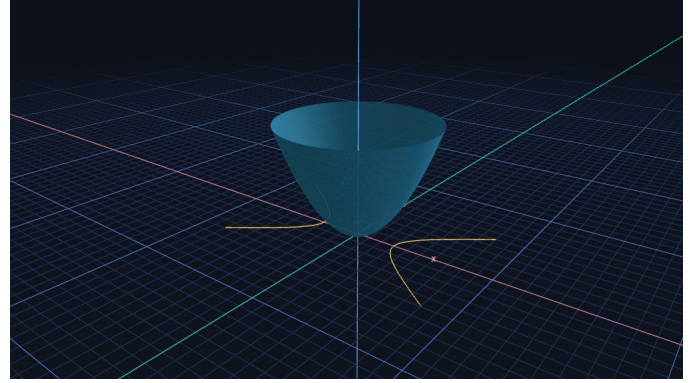


Figure 2. A shape on the plane in Analytic Geometry.

Alternatively, when selecting the Physics module, the environment allows the simulation of objects and rigid solids, exposing their dynamic interactions in a three-dimensional geometric plane. This approach makes it possible to observe physical phenomena in controlled scenarios, as detailed in the situations described in the next section.

VI. DISCUSSION

During the development of the site, we discussed the need, on the part of higher-education students, for the use of similar platforms. In doing so, we came across the fact that GeoGebra had around 300,000 monthly downloads until some time ago [9].

To demonstrate the applicability of Geopsics, we considered that, in addition to developing the interface, we could verify its functionality through problems extracted from the books by Paulo Winterle [4] and Halliday [5]. For the tests, we chose to start with analytic geometry.

The first scenario analyzed consists of Exercise 8, item a, from the chapter on quadric surfaces (p. 227) [4], whose proposal requires identifying the surface S defined by $y^2 - 4z^2 - 2x = 0$ and graphically determining its intersection with the plane $\pi : x - 2 = 0$.

Traditionally, solving this problem requires students to perform the algebraic substitution of the plane into the surface equation and to attempt to sketch a resulting conic on a two-dimensional plane (board or paper), omitting the spatial richness of the complete three-dimensional structure. By entering the equations directly into the Geopsics environment, the system performs the processing in real time through the core based on the Math.js library, generating the visualization shown in Figure 3.

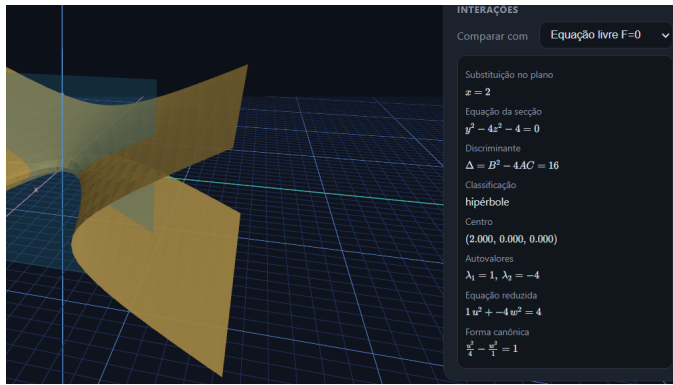


Figure 3. Opposite perspective along x .

As observed in the graphical interface of the tool (Figure 3), the analytic engine of Geopsics provides the user not only with the volumetric representation of the saddle and the intercepting plane, but also with the step-by-step algebraic breakdown in the side interaction panel. The software performs the parameterized substitution ($x = 2$), resulting in the section equation $y^2 - 4z^2 - 4 = 0$.

The ability to manipulate the plane in real time and observe the change in the resulting hyperbola mitigates the barrier of pure abstraction, allowing students to consolidate the spatial intuition required for subsequent disciplines, such as Differential and Integral Calculus.

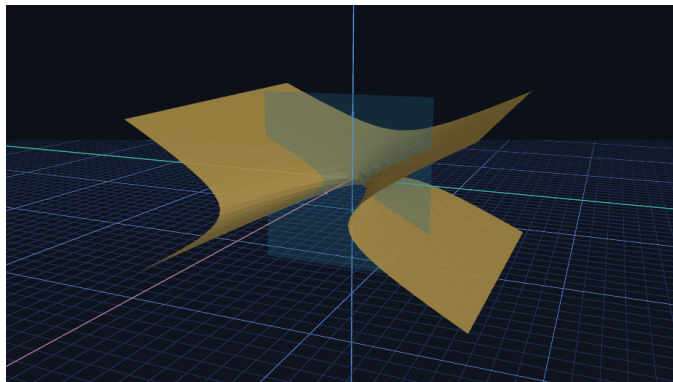


Figure 4. Angled perspective (x, y, z) .

Now, from the standpoint of Classical Physics, we selected Example 5:11 from the chapter on Applications of Newton's Laws in Sears and Zemansky (Young and Freedman): Physics I, which addresses the fundamental quantities of Uniform Circular Motion (UCM) from a dynamic perspective [10].

The problem proposes the scenario of an object with mass $m = 25$ kg at rest on an ideal ice surface (perfectly smooth and frictionless), attached to a fixed post by a rope of length $R = 5$ m. Once set in motion, the object begins to perform a uniform rotational motion, completing five revolutions per minute (5 rpm). The proposal requires determining the magnitude of the force F exerted by the rope on the object to keep it on this circular trajectory.

In this example, the student must manipulate kinematic equations before being able to run the simulation in Geopsics. The formal algebraic development is configured through the following fundamental steps. First, the period (T) is determined, representing the time interval required for the body to complete one full circular revolution. Knowing that the object performs 5 turns every 60 seconds:

$$T = \frac{60 \text{ s}}{5} = 12 \text{ s}$$

From the period, the angular velocity (ω) of the system is extracted in radians per second:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{12} = \frac{\pi}{6} \text{ rad/s} \approx 0,524 \text{ rad/s}$$

With the angular velocity established, the magnitude of the tangential linear velocity (v) of the body at the end of the 5 m radius is calculated:

$$v = \omega \cdot R = \frac{\pi}{6} \cdot 5 = \frac{5\pi}{6} \text{ m/s} \approx 2,62 \text{ m/s}$$

The continuous variation in the direction of the linear velocity vector generates a radial acceleration directed toward the center of the trajectory, called centripetal acceleration (a_c), calculated by:

$$a_c = \frac{v^2}{R} = \frac{\left(\frac{5\pi}{6}\right)^2}{5} = \frac{25\pi^2}{36 \cdot 5} = \frac{5\pi^2}{36} \text{ m/s}^2 \approx 1,37 \text{ m/s}^2$$

Finally, applying Newton's Second Law to curvilinear motion, the resultant centripetal force (F_c), which in this scenario is exerted exclusively by the tension of the rope (F), is given by the product of the mass of the body and its directed acceleration:

$$F = F_c = m \cdot a_c = 25 \cdot \left(\frac{5\pi^2}{36}\right) = \frac{125\pi^2}{36} \text{ N} \approx 34,27 \text{ N}$$

In the traditional teaching method, this abstract and purely operational path often prevents students from understanding the physics underlying the phenomenon. However, visualizing the dynamic and vector behavior surpasses the method that confines students to the "mere substitution of formulas on paper" [1], [11], [12]. The literature corroborates that such visualization and interaction accelerate the understanding of the mechanical phenomenon and foster student engagement and investigative autonomy during the transition to higher education.

Nevertheless, once these data are obtained, it is possible to transpose this scenario into Geopsics. The student begins to interact with the Mechanical Physics module by selecting the option "Centripetal (rotation)". To feed the platform's graphical and computational engine, the input parameters are filled in with the data obtained and converted in the analysis: the radius of the trajectory ($R = 5$ m), the calculated centripetal force ($F_c = 34,27$ N), and the mass of the object adjusted to the interface scale ($m = 25.000$ g).

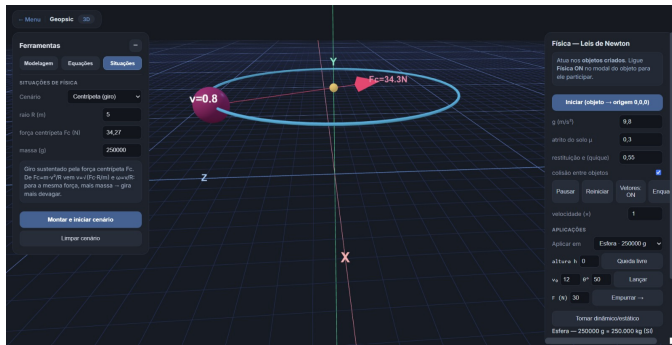


Figure 5. Simulation in Geopsics.

When processing the entered data, the computational core of Geopsics instantly renders the three-dimensional simulation of the system in motion. The platform dynamically displays the body describing the circular orbit around the post, but the main cognitive gain lies in the plotting and coupling of the vectors in real time: the linear velocity vector ($\vec{v}$) positioned tangentially to the trajectory and the centripetal force vector ($\vec{F}_c$) pointing radially toward the geometric center of the rotation.

From the perspective of Raymond Duval's registers of semi-otic representation [2], Geopsics acts efficiently by promoting the congruence between the algebraic register (Newton's equations and the parametric constraints provided by the problem statement) and the visual-geometric register (the kinematic behavior and the orientation of the vectors on the screen). This immediate and interactive transition stimulates the student's mechanical intuition, converting abstract concepts of vectors and forces into concrete spatial perceptions and facilitating the process of meaningful learning in fundamental physics disciplines.

VII. CONCLUSION

Therefore, since the obstacles in learning Analytic Geometry and Mechanical Physics are associated with the difficulty of transitioning between algebraic formulas and spatial geometric interpretation, this work contributes to this scenario by proposing a three-dimensional virtual environment that integrates both disciplines into a single simulation interface. The unification of these concepts helps students overcome the barrier of theoretical abstraction through direct observation, attacking the root of a pedagogical problem that historically results in high rates of retention and failure in university classrooms.

From the methodological standpoint, this work offers teachers an adaptable didactic resource that can be integrated both into lectures and into practical laboratory activities. By allowing the interactive manipulation of variables in linear equations, quadric surfaces, and the laws of classical mechanics, the software acts as a link between textbook theory and active experimentation. This flexibility supports teachers in following the students' resolution steps and in evaluating the class's performance in a continuous, direct, and transparent manner, making it simpler to diagnose specific conceptual gaps throughout the semester.

From the students' perspective, the platform's potential lies in fostering autonomy and the development of more complex cognitive skills. The need to configure and test physical and geometric parameters in the simulator, using the formulas as parameters, induces students to adopt an investigative posture, favoring metacognition. By confronting the generated visual results with their own abstract expectations, students are led to analyze their particular difficulties in spatial visualization and to better understand the purpose of the mathematical tools they used. This mathematical self-experience enriches both individual study and collective classroom discussions, providing a practical parameter for discussing the learning process at the undergraduate level.

In this way, the incorporation of interactive technologies is consolidated as a viable alternative for modernizing teaching methodologies in the exact sciences. Although the planning and implementation of these tools require dedication and the overcoming of technical and infrastructure challenges on the part of the teaching staff [13], the pedagogical benefits justify the institutional investment. As future developments of this work, we suggest the regular application of the software in undergraduate classes in order to collect qualitative and quantitative data that allow measuring the tool's real impact on long-term academic performance.

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